

Chapter 8 Hamiltonian and Hamilton's Eqns of Motion

$$L = T - V = L(q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \Big|_{q_j, \dot{q}_j} \right) - \frac{\partial L}{\partial q_i} \Big|_{q_j, \dot{q}_j}, \quad p_i = \frac{\partial L}{\partial \dot{q}_i} \Big|_{q_j, \dot{q}_j}. \quad (1)$$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

$$L = L(r, \dot{r}, p_\theta)$$

~~$\Downarrow$ Lagrange Eqn~~

Legendre transformation

$$q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n \longrightarrow q_1 \dots q_n, p_1 \dots p_n$$

$$f(x, y) \implies df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

In thermodynamics

$$\text{1st Law} \quad dU = dQ + dW = \overset{\text{fluid at constant } N}{T dS - p dV}, \quad U = U(S, V) \text{ internal energy}$$

We want T, V as variables.

$$F = A = U - TS \quad \text{Helmholtz free energy}$$

$$dA = dU - d(TS) = T dS - p dV - T dS - S dT = -S dT - p dV \implies A = A(T, V).$$

In general

$$f(x, y) = u dx + v dy, \quad u = \frac{\partial f}{\partial x}, \quad v = \frac{\partial f}{\partial y}, \quad (x, y) \longrightarrow (u, v)$$

$$g = f - ux \implies dg = df - d(ux) = \cancel{u dx} + v dy - \cancel{u dx} - x du = -x du + v dy, \quad g = g(u, v).$$

Let us look at $L(q_1 \dots q_n, \dot{q}_1 \dots \dot{q}_n)$, with

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \Big|_{q_j, \dot{q}_j}$$

$$H = - \left(L - \sum_i p_i \dot{q}_i \right)$$

$$\begin{aligned} dH &= -dL + \sum_i d(p_i \dot{q}_i) \\ &= - \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i} \right) - \frac{\partial L}{\partial t} dt + \sum_i (p_i \cancel{d\dot{q}_i} + \dot{q}_i dp_i) \end{aligned}$$

so that

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right) \implies H = H(p_1 \dots p_n, q_1 \dots q_n, t).$$

$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} \stackrel{\text{Eq. (1)}}{=} -\dot{p}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$
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CANONICAL EQS. OF MOTION